

MSE 311 Thermodynamics of materials-Homework 10

Name:

Student ID:

Please hand in this homework before class on November 24th.

1. Derive the following formula from Gibbs-Duhem equation.

$$\Delta G^M = RT \left[X_A \ln X_A + (1 - X_A) \ln (1 - X_A) \right]$$

2. Derive the following formula

$$X_A d \ln \gamma_A + X_B d \ln \gamma_B = 0$$

3. Calculate the heat required to form a liquid solution at 1356 K, starting with 1 mole of Cu and 1 mole of Ag at 298 K. At 1356 K, the molar heat of mixing of liquid Cu and liquid Ag is given by $\Delta H^M = -20,590 X_{\text{Cu}}X_{\text{Ag}}$.

4. Melts in the system Pb–Sn exhibit regular solution behavior. At 473°C, $a_{\text{Pb}} = 0.055$ in a liquid solution of $X_{\text{Pb}} = 0.1$. Calculate the value of interaction parameter Ω for the system and calculate the activity of Sn in the liquid solution of $X_{\text{Sn}} = 0.5$ at 500°C.

$$\gamma_{\text{Pb}} = \frac{a_{\text{Pb}}}{X_{\text{Pb}}} = \frac{0.055}{0.1} = 0.55$$

$$RT \ln \gamma_{\text{Pb}} = \Omega(1 - X_{\text{Pb}})^2$$

$$\Omega = -4578J$$

$$RT \ln \gamma_{\text{Sn}} = \Omega(1 - X_{\text{Sn}})^2$$

$$\gamma_{\text{Sn}} = 0.837$$

$$a_{\text{Sn}} = \gamma_{\text{Sn}} X_{\text{Sn}} = 0.837 \times 0.5 = 0.418$$

1. Using the virial equation of state for hydrogen at 298 K is $PV=RT(1 + 6.4 \times 10^{-4}P)$, calculate
 - a. The fugacity of hydrogen at 500 atm and 298 K,
 - b. The pressure at which the fugacity is twice the pressure,
 - c. The change in the Gibbs free energy caused by a compression of 1 mole of hydrogen at 298 K from 1 to 500 atm
 - d. What is the magnitude of the contribution to (c) caused by the nonideality of hydrogen?

$$\begin{aligned}
 a. \quad V &= \frac{RT}{P} + 6.4 \times 10^{-4} RT \Rightarrow \alpha = -6.4 \times 10^{-4} RT \\
 f &= Pe^{-\alpha P/RT} = 500 \times e^{6.4 \times 10^{-4} \times 500} = 689 \text{ atm} \\
 b. \quad \frac{f}{P} &= e^{-\alpha P/RT} = 2 \Rightarrow P = -\frac{RT \ln 2}{\alpha} = \frac{\ln 2}{6.4 \times 10^{-4}} = 1083 \text{ atm} \\
 dG &= -SdT + VdP = \left(\frac{RT}{P} + 6.4 \times 10^{-4} RT \right) dP \\
 c. \quad \Delta G &= \int_{P_1}^{P_2} \left(\frac{RT}{P} + 6.4 \times 10^{-4} RT \right) dP = RT \ln \left(\frac{P_2}{P_1} \right) + 6.4 \times 10^{-4} RT \Delta P = 16188 J \\
 d. \quad \Delta G &= 6.4 \times 10^{-4} RT \Delta P = 791 J
 \end{aligned}$$

2. Considering the activity coefficients of the components of a binary solution could be represented by power series of the form

$$\begin{aligned}
 \ln \gamma_A &= \alpha_1 X_B + \frac{1}{2} \alpha_2 X_B^2 + \frac{1}{3} \alpha_3 X_B^3 + \frac{1}{4} \alpha_4 X_B^4 + \dots \\
 \ln \gamma_B &= \beta_1 X_A + \frac{1}{2} \beta_2 X_A^2 + \frac{1}{3} \beta_3 X_A^3 + \frac{1}{4} \beta_4 X_A^4 + \dots
 \end{aligned}$$

Are α_3 and β_3 equal to zero? What about α_4 and β_4 ? Why?

$$\begin{cases} d \ln \gamma_A = \alpha_1 dX_B + \alpha_2 X_B dX_B + \alpha_3 X_B^2 dX_B + \alpha_4 X_B^3 dX_B \\ d \ln \gamma_B = \beta_1 dX_A + \beta_2 X_A dX_A + \beta_3 X_A^2 dX_A + \beta_4 X_A^3 dX_A \end{cases}$$

Gibbs-Duhem equation:

$$\begin{aligned}
 X_A d \ln \gamma_A + X_B d \ln \gamma_B &= 0 \\
 X_A (\alpha_1 + \alpha_2 X_B + \alpha_3 X_B^2 + \alpha_4 X_B^3) dX_B + X_B (\beta_1 + \beta_2 X_A + \beta_3 X_A^2 + \beta_4 X_A^3) dX_A &= 0 \\
 \alpha_1 X_A + \beta_1 X_B + X_A X_B (\alpha_2 + \beta_2 + \alpha_3 X_B + \beta_3 X_A + \alpha_4 X_B^2 + \beta_4 X_A^2) &= 0
 \end{aligned}$$

According to $X_A + X_B = 1$, then

$$\begin{aligned}
 dX_A + dX_B &= 0 \Rightarrow dX_A = -dX_B \\
 X_A (\alpha_1 + \alpha_2 X_B + \alpha_3 X_B^2 + \alpha_4 X_B^3) - X_B (\beta_1 + \beta_2 X_A + \beta_3 X_A^2 + \beta_4 X_A^3) &= 0 \\
 \alpha_1 X_A - \beta_1 X_B + X_A X_B (\alpha_2 - \beta_2 + \alpha_3 X_B - \beta_3 X_A + \alpha_4 X_B^2 - \beta_4 X_A^2) &= 0 \\
 X_B &= (1 - X_A) \\
 X_A (\alpha_1 + \alpha_2 (1 - X_A) + \alpha_3 (1 - X_A)^2 + \alpha_4 (1 - X_A)^3) - (1 - X_A) (\beta_1 + \beta_2 X_A + \beta_3 X_A^2 + \beta_4 X_A^3) &= 0 \\
 = -\beta_1 + (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \beta_2 + \beta_1) X_A + (-\alpha_2 - 2\alpha_3 - 3\alpha_4 - \beta_3 + \beta_2) X_A^2 + (\alpha_3 + 3\alpha_4 - \beta_4 + \beta_3) X_A^3 + (-\alpha_4 + \beta_4) X_A^4 &= 0 \\
 \begin{cases} -\beta_1 = 0 \\ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \beta_2 + \beta_1 = 0 \\ -\alpha_2 - 2\alpha_3 - 3\alpha_4 - \beta_3 + \beta_2 = 0 \\ \alpha_3 + 3\alpha_4 - \beta_4 + \beta_3 = 0 \\ -\alpha_4 + \beta_4 = 0 \end{cases} \Rightarrow \begin{cases} \alpha_1 = \beta_1 = 0 \\ \alpha_2 = \beta_2 \\ \alpha_3 = \beta_3 = -\alpha_4 = -\beta_4 \end{cases}
 \end{aligned}$$

3. Melts in the system Pb-Sn exhibit regular solution behavior. At 473°C, $a_{Pb} = 0.055$ in a liquid

solution of $X_{Pb} = 0.1$. Calculate the value of interaction parameter Ω for the system and calculate the activity of Sn in the liquid solution of $X_{Sn} = 0.5$ at 500°C.

$$\gamma_{Pb} = \frac{a_{Pb}}{X_{Pb}} = \frac{0.055}{0.1} = 0.55$$

$$RT \ln \gamma_{Pb} = \Omega(1 - X_{Pb})^2$$

$$\Omega = \frac{RT \ln \gamma_{Pb}}{(1 - X_{Pb})^2} = \frac{8.314 \times 746 \times \ln 0.55}{(1 - 0.1)^2} = -4578 J$$

$$RT \ln \gamma_{Sn} = \Omega(1 - X_{Sn})^2$$

$$\gamma_{Sn} = e^{\frac{\Omega(1 - X_{Sn})^2}{RT}} = e^{\frac{-4578 \times (1 - 0.5)^2}{8.314 \times 773}} = 0.837$$

$$a_{Sn} = \gamma_{Sn} X_{Sn} = 0.837 \times 0.5 = 0.418$$

4. Calculate the heat required to form a liquid solution at 1356 K, starting with 1 mole of Cu and 1 mole of Ag at 298 K. At 1356 K, the molar heat of mixing of liquid Cu and liquid Ag is given by $\Delta H^M = -20,590 X_{Cu}X_{Ag}$.

Hint: The total heat includes the heat required to increase temperatures of each components and the heat required due to the mixing of two components.

$$c_{p,Cu(s)} = 23.0 + \frac{1 \times 10^5}{T^2} \quad [J/mol/K]$$

$$c_{p,Cu(l)} = 24.0 \quad [J/mol/K]$$

$$c_{p,Ag(s)} = 21.0 + 9 \times 10^{-3} T + \frac{2 \times 10^5}{T^2} \quad [J/mol/K]$$

$$c_{p,Ag(l)} = 30.0 \quad [J/mol/K]$$

$$\Delta H_{melt,Cu,1356K} = 14000 \quad [J/mol]$$

$$\Delta H_{melt,Ag,1234K} = 12000 \quad [J/mol]$$

$$Q_{Cu} = \int_{298}^{1356} c_{p,Cu(s)} dT + \Delta H_{melt,Cu,1356K} = 23.0 \times (1356 - 298) - \frac{1 \times 10^5}{1356} + \frac{1 \times 10^5}{298} + 14000 = 38596 J / mol$$

$$Q_{Ag} = \int_{298}^{1234} c_{p,Ag(s)} dT + \int_{1234}^{1356} c_{p,Ag(l)} dT + \Delta H_{melt,Ag,1234K} = 21.0 \times (1234 - 298) + \frac{1}{2} \times 9 \times 10^{-3} \times (1234^2 - 298^2) - \frac{2 \times 10^5}{1234} + \frac{2 \times 10^5}{298}$$

$$\Delta H^M = -20590 X_{Cu} X_{Ag} = -20590 \times 0.5 \times 0.5 = -5148 J / mol$$

$$Q = Q_{Cu} + Q_{Ag} + 2\Delta H^M = 38596 + 42278 - 2 \times 5148 = 70578 J$$